Configuração 6: $M = 128$, $C = 8$, $K = 8$, IoT = 20 dB, QPSK
Perda de SNR do BCD-MMSE (1 iteração) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas
Fundo laranja no rótulo: o modo com o melhor resultado no par, comparando do formato com menos bits para o de mais, sem contar as colunas em que todos os modos ficam abaixo de 0,05 dB
Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo

Perda de SNR (dB)

< 0,1 dB | 0,1 a 0,5 dB | 0,5 a 1 dB | 1 a 2 dB | 2 a 3 dB | 3 a 5 dB | 5 a 10 dB | > 10 dB | × não chega a 10^{-2} | rótulo do melhor modo

TDLA30-5

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.58 1.8%	0.55 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.58 1.8%	0.43 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	1.58 1.9%	0.55 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	1.58 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	1.58 1.8%	0.64 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	1.52 2.0%	0.55 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.52 1.8%	0.43 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.52 2.2%	0.55 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.52 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.52 2.0%	0.63 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.62 2.0%	0.55 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.52 1.8%	0.43 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.62 2.1%	0.55 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.7%	0.01 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.70 2.0%	0.65 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.62 2.0%	0.55 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.52 1.8%	0.43 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.62 2.2%	0.55 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.6%	0.01 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.70 2.0%	0.65 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.62 1.9%	0.55 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.52 1.8%	0.43 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.62 2.1%	0.55 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.8%	0.01 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.71 2.0%	0.65 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.62 2.0%	0.55 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.52 1.8%	0.43 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.62 2.2%	0.55 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.6%	0.01 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.71 2.0%	0.65 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.62 2.0%	0.55 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.52 1.8%	0.43 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.62 2.2%	0.55 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.8%	0.01 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.71 2.0%	0.65 2.4%	0.00 3.7%	0.00 4.2%

TDLA30-75

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.82 1.8%	0.56 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.82 1.8%	0.40 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	1.82 1.9%	0.56 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	1.82 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	1.82 1.8%	0.62 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	1.48 2.0%	0.56 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.48 1.8%	0.41 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.48 2.2%	0.56 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.48 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.48 2.0%	0.61 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.63 2.0%	0.57 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.53 1.8%	0.40 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	0.63 2.1%	0.57 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	0.67 2.0%	0.61 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.63 2.0%	0.57 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.52 1.8%	0.40 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	0.63 2.2%	0.57 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.6%	0.01 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	0.67 2.0%	0.61 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.63 1.9%	0.57 2.2%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.52 1.8%	0.40 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	0.63 2.1%	0.57 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.8%	0.01 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	0.67 2.0%	0.61 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.63 2.0%	0.57 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.53 1.8%	0.40 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	0.63 2.2%	0.57 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.6%	0.01 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	0.67 2.0%	0.61 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.63 2.0%	0.57 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.53 1.8%	0.40 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	0.63 2.2%	0.57 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	0.20 1.8%	0.01 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	0.67 2.0%	0.61 2.4%	0.01 3.7%	0.00 4.2%

TDLA30-300

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.80 1.8%	0.62 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.80 1.8%	0.39 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	1.80 1.9%	0.61 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	1.80 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	1.80 1.8%	0.57 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	1.66 2.0%	0.61 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.66 1.8%	0.38 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.66 2.2%	0.60 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.66 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.66 2.0%	0.55 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.67 2.0%	0.61 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.54 1.8%	0.38 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.67 2.1%	0.61 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.22 1.7%	0.01 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.62 2.0%	0.57 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.68 2.0%	0.61 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.55 1.8%	0.38 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.68 2.2%	0.61 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.22 1.6%	0.01 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.62 2.0%	0.57 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.68 1.9%	0.61 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.55 1.8%	0.38 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.68 2.1%	0.61 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.22 1.8%	0.01 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.62 2.0%	0.57 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.67 2.0%	0.61 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.54 1.8%	0.38 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.67 2.2%	0.61 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.22 1.6%	0.01 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.62 2.0%	0.57 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.68 2.0%	0.61 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.55 1.8%	0.38 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.67 2.2%	0.61 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.22 1.8%	0.01 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.62 2.0%	0.57 2.4%	0.00 3.7%	0.00 4.2%

TDLC300-100

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.76 1.8%	0.44 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.76 1.8%	0.51 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.76 1.9%	0.44 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.76 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	1.76 1.8%	0.60 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	1.84 2.0%	0.45 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.84 1.8%	0.53 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.84 2.2%	0.45 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.84 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.84 2.0%	0.60 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.57 2.0%	0.44 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.63 1.8%	0.53 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.57 2.1%	0.44 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.25 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.69 2.0%	0.61 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.57 2.0%	0.44 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.63 1.8%	0.53 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.57 2.2%	0.44 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.25 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.69 2.0%	0.61 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.57 1.9%	0.44 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.63 1.8%	0.53 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.57 2.1%	0.44 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.25 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.69 2.0%	0.61 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.57 2.0%	0.44 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.63 1.8%	0.53 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.57 2.2%	0.44 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.25 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.69 2.0%	0.61 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.56 2.0%	0.44 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.62 1.8%	0.53 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.56 2.2%	0.44 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.25 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.69 2.0%	0.61 2.4%	0.00 3.7%	0.00 4.2%

TDLB100-400

Quantização	Estouro	<22,10>	<26,10>	<32,12>	<36,12>
AP_TRN	AP_SAT	2.89 1.7%	2.84 2.0%	0.00 3.3%	0.00 3.7%
	AP_SAT_ZERO	3.58 1.6%	3.55 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	2.89 1.8%	2.84 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP	2.91 1.6%	3.05 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	3.61 1.7%	3.57 2.2%	0.00 3.9%	0.00 4.4%
AP_TRN_ZERO	AP_SAT	3.28 1.9%	2.84 2.3%	0.01 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.70 1.7%	3.62 2.2%	0.01 3.9%	0.00 4.4%
	AP_SAT_SYM	3.28 2.1%	2.84 2.5%	0.01 3.9%	0.00 4.3%
	AP_WRAP	3.35 1.4%	3.07 1.9%	0.01 3.4%	0.00 3.9%
	AP_WRAP_SM	3.75 1.9%	3.59 2.3%	0.01 3.9%	0.00 4.4%
AP_RND	AP_SAT	2.85 1.8%	2.84 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.49 1.7%	3.60 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.85 2.0%	2.84 2.4%	0.00 3.9%	0.00 4.4%
	AP_WRAP	3.02 1.6%	3.05 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	3.55 1.9%	3.53 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_ZERO	AP_SAT	2.85 1.9%	2.84 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.49 1.7%	3.60 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.85 2.1%	2.84 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	3.02 1.4%	3.05 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	3.56 1.9%	3.53 2.3%	0.00 3.9%	0.00 4.4%
AP_RND_MIN_INF	AP_SAT	2.85 1.8%	2.84 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.49 1.7%	3.60 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	2.85 2.0%	2.84 2.4%	0.00 3.9%	0.00 4.3%
	AP_WRAP	3.02 1.6%	3.05 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	3.56 1.9%	3.53 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_INF	AP_SAT	2.85 1.9%	2.84 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.49 1.7%	3.60 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.85 2.1%	2.84 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	3.02 1.4%	3.05 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	3.55 1.9%	3.53 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_CONV	AP_SAT	2.85 1.9%	2.84 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	3.49 1.7%	3.60 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.85 2.1%	2.84 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	3.02 1.6%	3.05 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	3.55 1.9%	3.53 2.2%	0.00 3.9%	0.00 4.4%